

Applications in Consumer Behavior and Inertia

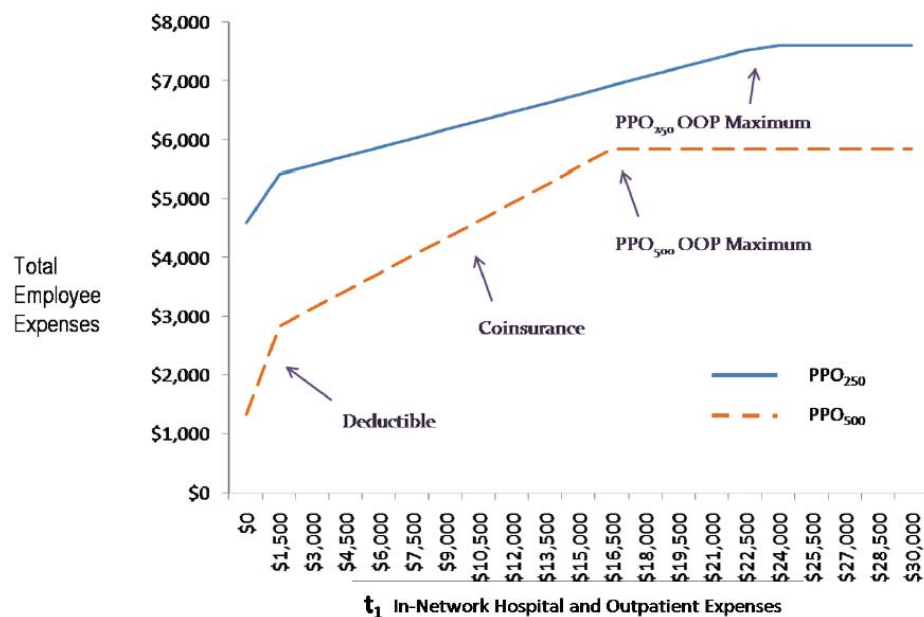
Econ 649

Public Policies where Consumers Make Important Choices (but didn't in past)

- Prior regime: good where consumers by default “purchased” a product that someone else chose
- New regime: consumers “choose” their preferred product
- Examples of “consumer choice” programs
 - Health insurance options (Handel)
 - Privatized social security (Duarte & Hastings)
 - Retail competition in residential electricity (Hortacsu, Madanizadeh, and Puller)
 - Grade schools (Hastings, Kane and Staiger)

In New Choice Programs, Some Consumers Make “Unusual” Choices

- Hastings, Kane and Staiger: parents of kids in neighborhoods do not send kids to better schools when given the option
- Handel: employees given a new health insurance option “choose” the more expensive one



many who were enrolled in PPO250 and became dominated by PPO500, stayed in PPO250

In New Choice Programs, Some Consumers Make “Unusual” Choices

- Hortacsu, Madanizadeh and Puller – homeowners can spend a few minutes to choose a cheaper electricity provider to provide the same power but they don’t
- Hastings, Hortacsu and Syverson – employees pay “too much” in fees when paying for privatized social security
- Allcott and Wozny – car purchasers don’t fully account for effect of better MPG on cost savings of using a car

All of these examples of inertia have policy implications!

In New Choice Programs, Some Consumers Make “Unusual” Choices

- The “reduced-form” descriptive analysis of these is “Stupid Human Tricks with conjecture”
- But understanding mechanism is important for making policy prescriptions
 - Are consumers uninformed?
 - Are search costs high?
 - Are switching costs high?
 - Do they discount “properly”?
 - Do they value product characteristics that policymakers may not have anticipated?
- To understand the mechanism, we often need to put more structure on empirical approach

Adverse Selection and Inertia in Health Insurance Markets: When Nudging Hurts

Ben Handel

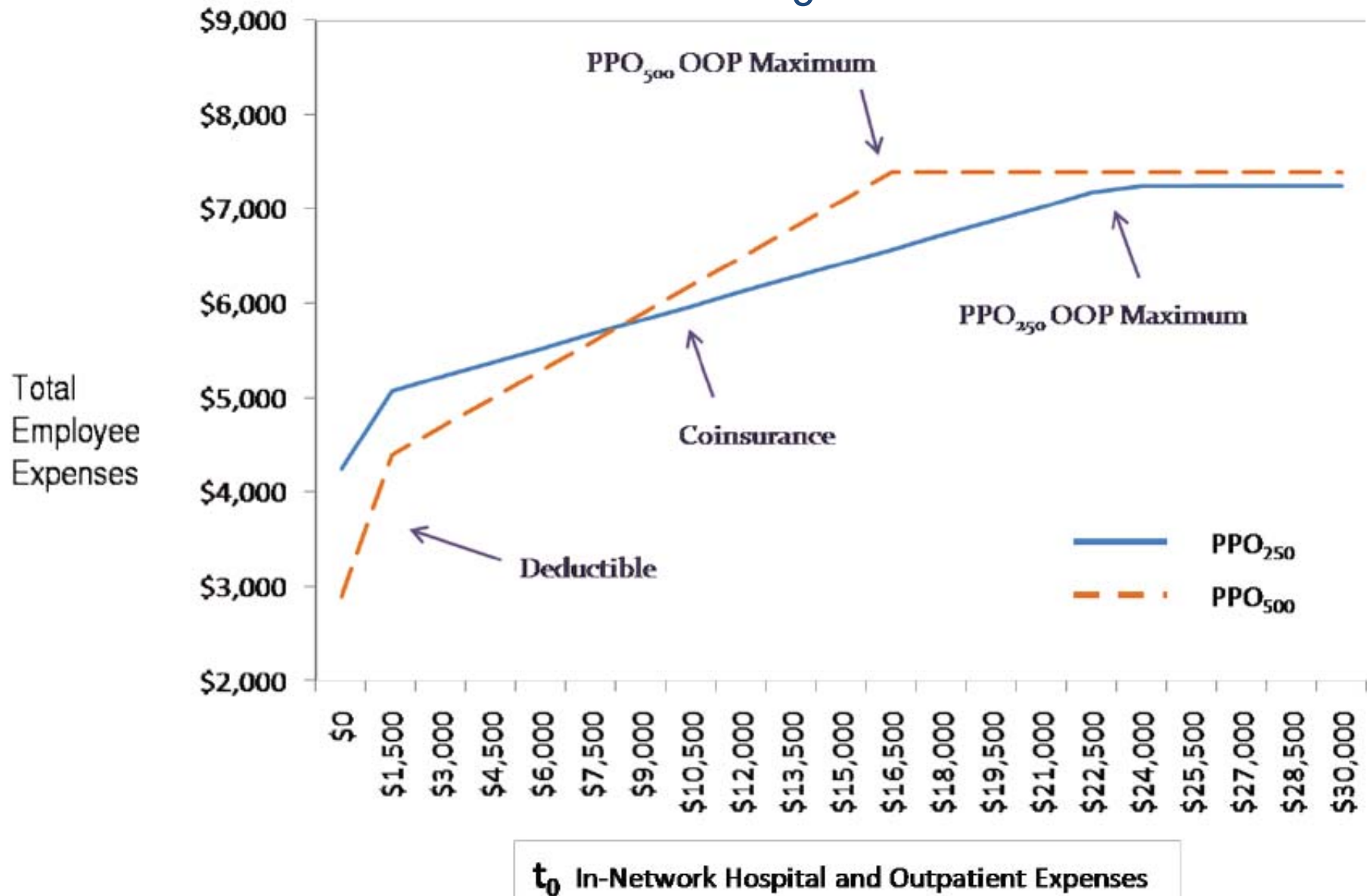
Relationship to Existing Literature

- Adverse Selection in insurance:
 - When all risks cannot be observed & priced, riskier consumers choose comprehensive coverage, which raises prices and induces less risky consumers to choose less comprehensive plan, which raises prices in comprehensive plan which...
 - Inefficiency: healthy but risk averse will not purchase (enough) insurance
- Consumer inertia present
- Handel: Interaction?
 - What if reducing inertia (changing demand) affects pricing on the supply side? What are the consumer welfare effects?
 - Can make consumers worse off.
 - Intuition: when switching costs removed, the most comprehensive health plan unravels much quicker (and prices rise b/c only sickest patients in comprehensive plan)

Institutional Setting

- Large firm changed health plan menu; existing employees had to choose plan (no default)
 - Subsequently, default was current plan
- 3 plan options
 - Same medical services (providers, services)
 - Different combo of low (high) premiums and high (low) cost sharing

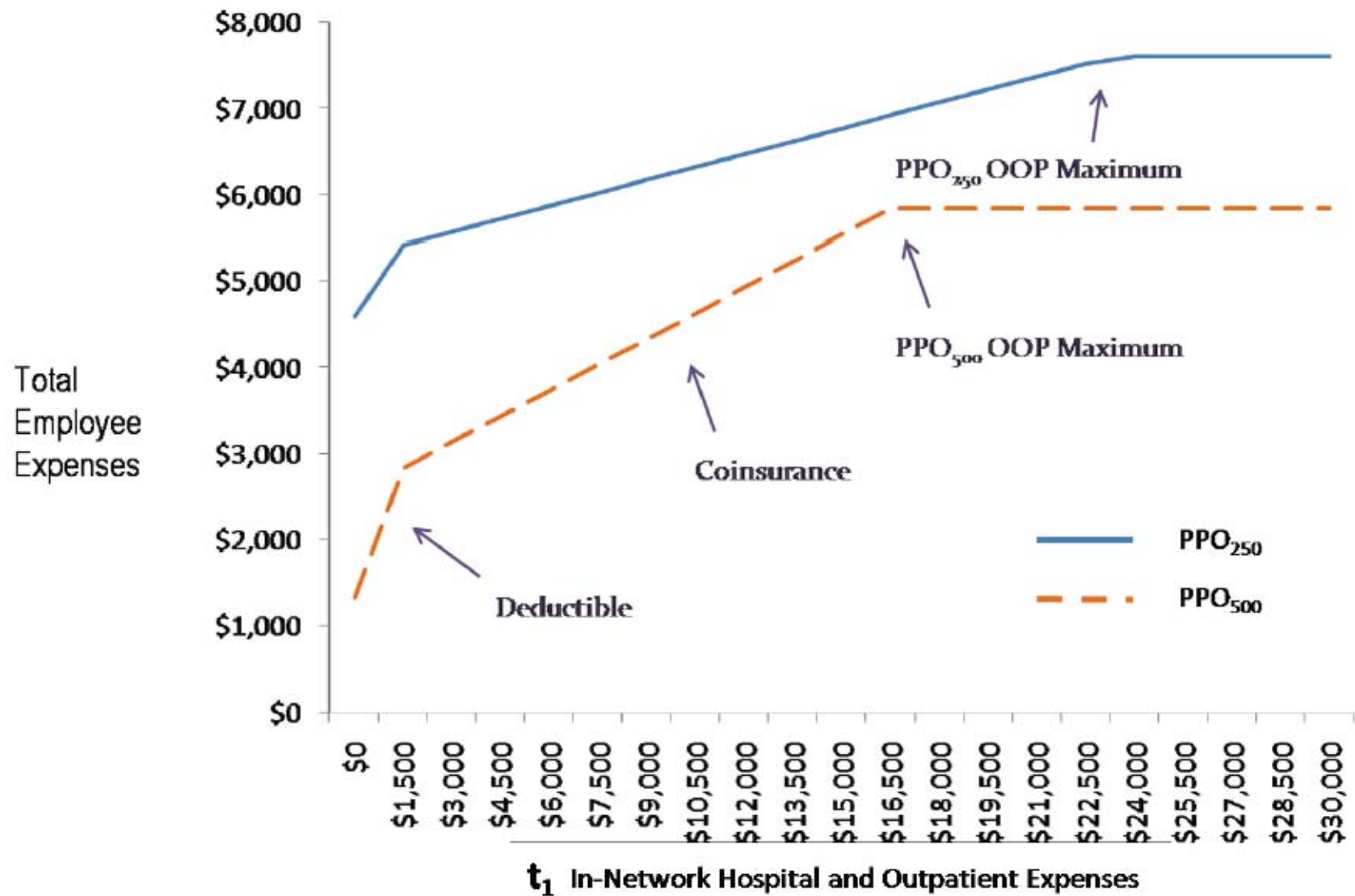
PERIOD t_0



Note: for low income families

Ex post, “healthy” should choose 500 while “sick” should choose 250

PERIOD t_1



Ex post, everyone should choose 500

Components Need to Model

- Goal: Counterfactual
“experiments”/calculations: what if switching costs fell?
- Demand-side:
 - Switching costs
 - Expected health expenditures
 - Risk preferences
- Supply-side:
 - “Competitive”
 - Prices covers last year’s average cost, given mix of enrollees

Data

- For all 9000 employees:
 - Lots of demographics
 - Plan choice
 - Claims
 - Use Johns Hopkins model to predict distribution of expenditures (assume rational expectations)
- At date t_0 , firm switched to offering 2 HMOs and 3 PPOs
 - Differed only in deductible, coinsurance, and out-of-pocket maximum
 - No default; had to choose
 - Not much switching b/t HMO & PPO, so analysis on PPO choice

Suggestive Evidence of Switching Costs

1. New employees (with no switching costs) choose different plans from (demographically similar) existing employees with switching costs
2. When one plan becomes strictly dominated, majority of consumers continue to “choose” it

Suggestive Evidence #1

New Enrollee Analysis			
	New Enrollee t_{-1}	New Enrollee t_0	New Enrollee t_1
N, t_0	1056	1377	-
N, t_1	784	1267	1305
t_0 Choices			
PPO_{250}	259 (25%)	287 (21%)	-
PPO_{500}	205 (19%)	306 (23%)	-
PPO_{1200}	155 (15%)	236 (17%)	-
HMO_1	238 (23%)	278 (20%)	-
HMO_2	199 (18%)	270 (19%)	-
t_1 Choices			
PPO_{250}	182 (23%)	253 (20%)	142 (11%)
PPO_{500}	201 (26%)	324 (26%)	562 (43%)
PPO_{1200}	95 (12%)	194 (15%)	188 (14%)
HMO_1	171 (22%)	257 (20%)	262 (20%)
HMO_2	135 (17%)	239 (19%)	151 (12%)
Demographics			
Mean Age	33	33	32
Median Age	31	31	31
Female %	56%	54%	53%
Manager %	20%	18%	19%
FSA Enroll %	15%	12%	14%
Dental Enroll %	88%	86%	86%
Median (Mean) Expense t_1	844 (4758)	899 (5723)	-
Income Tier 1	48%	50%	47%
Income Tier 2	33%	31%	32%
Income Tier 3	10%	10%	12%
Income Tier 4	5%	4%	4%
Income Tier 5	4%	5%	5%

Table 4: This table describes the choice behavior of new employees at the firm over several consecutive years and presents our first model-free test of switching costs. Each column describes one cohort of new employees at the firm, corresponding to a specific year of arrival. First, the chart describes the health insurance choices made by these cohorts in year t_0 (the year of the insurance plan menu change) and in the following year, t_1 . The last part of the chart lists the demographics for each cohort of new arrivals at the time of their arrival. Given the very similar demographic profiles and large sample size for each cohort, if there are no switching costs, the t_1 choices of employees who entered the firm at t_0 and t_{-1} should be identical to the t_1 choices of employees who entered the firm at t_1 . The table shows that, in fact, the active choices made by the t_1 cohort are quite different than those of the prior cohorts in the manner we would expect with high switching costs: the t_1 choices of employees who enter at t_0 and t_{-1} reflect both t_1 prices and t_0 choices while the t_1 choices of new employees at t_1 reflect t_1 prices. New employees at t_0 do not adjust to the significant price change from t_0 to t_1 while new employees' choices do reflect these price changes. This illustrates the large impact that switching costs have on choices in our setting, independent of the choice model setup and structure.

Good technique,
especially for a 65
page paper

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Must make active decision

New enrollees are demographically similar

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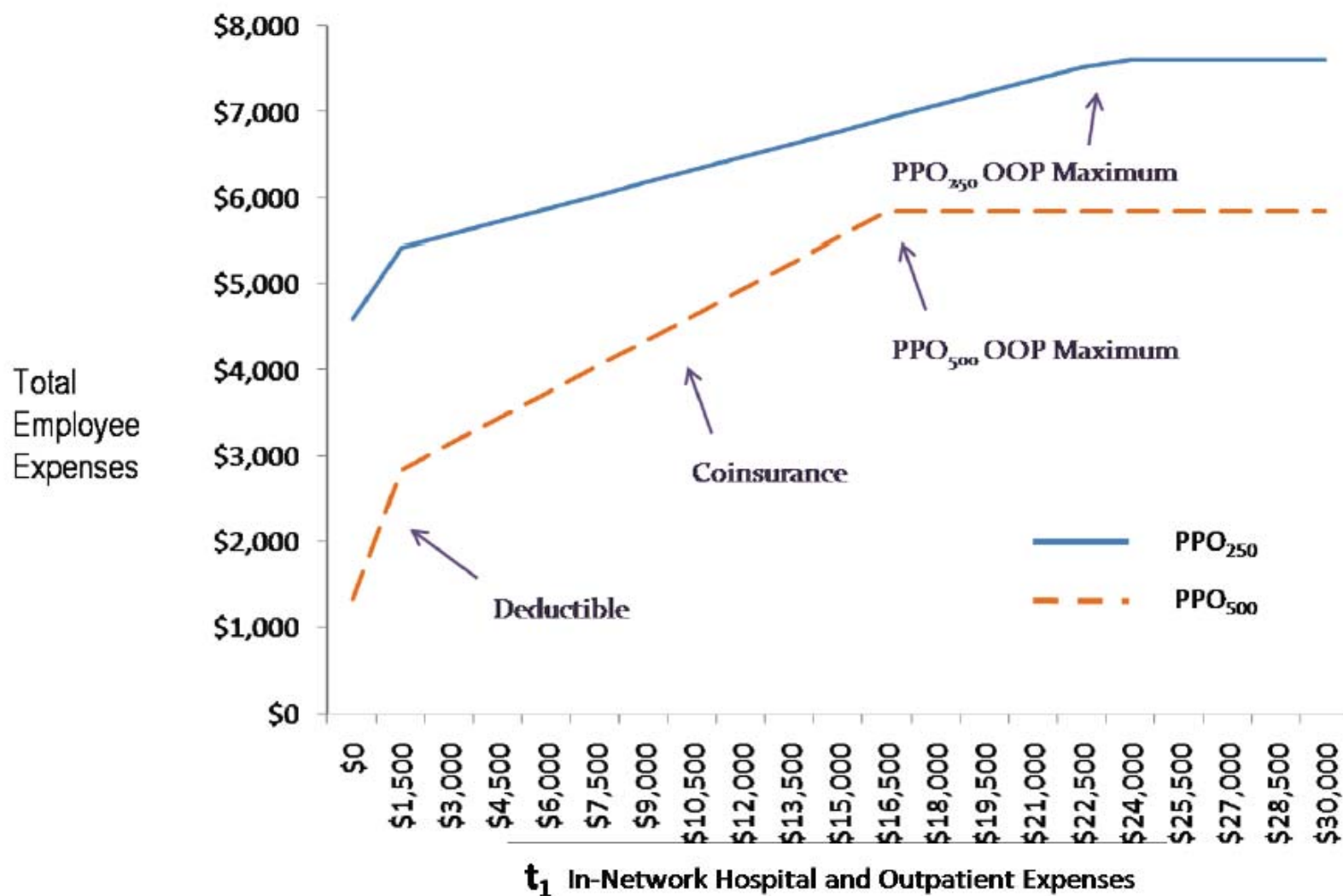
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← Should be same if no switching costs

But new enrollees (with no switching costs) much less likely to choose strictly dominated plan

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Suggestive Evidence #2



498 of 559 who were enrolled in PPO250 and became dominated by PPO500, stayed in PPO250

Value & Limitation of “Descriptive Evidence”

- Very credible evidence that switching costs exist and inhibit CS-improving choices
- But, does not tell us:
 - “welfare effect” of switching costs (or any meaningful estimate of magnitude)
 - How welfare would change if switching costs were reduced
- Of course to say more, need more assumptions

Model of Choice

$$U_{kjt} = \int_0^{\infty} f_{kjt}(OOP) u(OOP, P_{kjt}, \mathbf{1}_{kj,t-1}, W_k, X_k, Y_k, H_k) dOOP$$

- Recall, it's a homog good, so it's all about costs
- Expected utility for family k of choosing plan j in period t
- OOP = realization of medical expenses
- P_{kjt} = premium
- $\mathbf{1}_{kj,t-1}$ captures switching costs (enrolled in j in period $t-1$)
- $f_{kjt}(\cdot)$ is expense distribution
- Others capture wealth, income, category of dependents and family characteristics

Risk Aversion

$$u(x) = -\frac{1}{\gamma}e^{-\gamma x}$$

- x = consumption
- Constant Absolute Risk Aversion (CARA)
 - Bigger gamma more curved utility
→ more risk averse

- So for any realization of OOP:

$$u(\cdot) = \frac{-1}{\gamma_k(X_k)} e^{-\gamma_k(X_k)(W_k - P_{kjt} - OOP + \eta(Y_k)\mathbf{1}_{kj,t-1} + \delta_k(Y_k)\mathbf{1}_{1200} + \alpha_j(Y_k)H_k + \epsilon_{kjt})}$$

Switching cost (needs to prefer another plan by at least $\$ \eta$ to switch)

1200 has health Savings
Acct (product diff)

control for “high
cost” family

i.i.d.

Estimation

- Discrete choice at family level over sequence of decisions from t_0 to t_2
- Random coefficients probit via simulated ML

Should the demand side model be more dynamic?

- Some referees will likely think yes, others no
- Handel does not want to do this, saying:

First, price changes are not signaled in advance and change as a function of factors that would be difficult for consumers to model.³² Second, it is unlikely that most consumers can forecast substantial changes to their health status more than one year in advance. Third, in this empirical setting consumers make initial choices that make little sense in the context of a fully dynamic approach. They choose (and stay with) plans at t_0 that provide poor long run value given the time path of prices and health expectations. While it is clear that dynamics are an important component of choice with switching costs in some settings (e.g. cable provider or cell phone plan choice), in this setting we believe that incorporating empirically relevant dynamic aspects would not markedly change the results.

Identification

- Switching costs
 - Researcher can see periods with clearly ‘active’ and clearly ‘passive’ choices
 - At t_0 , no default, so no switching costs
 - At t_1 , there are switching costs
 - Prices change a lot in t_1
 - Should induce switching if low switching costs
 - (Cannot use new enrollees -- no switching costs -- b/c don't have claims history to estimate $f(.)$)
- Risk parameter
 - At t_0 there are no switching costs and only product differentiation is cost sharing under uncertain health outcomes

Estimates of Switching Costs

Empirical Model Results		
Parameter	Primary	Base
Switching Cost - Single, η_0	1729 (28)	1779 (72)
Switching Cost - Family, $\eta_0 + \eta_2$	2480 (26)	2354 (62)
SC - FSA Enroll, η_1	-551 (56)	- -
SC - Income, η_1	-32 (13)	- -
SC - Quantitative, η_1	5 (138)	- -
SC - Manager, η_1	198 (292)	- -
SC - Chronic Condition, η_1	80 (46)	- -
SC - Salient Change, η_1	156 (83)	- -
SC - PPO_{1200} , η_1	-19 (184)	- -
SC - Total Pop. Mean, η [Pop. Standard Deviation]	2032 [446]	2087 [286]

Literal interpretation:
those who previously have
made a choice forego about
\$2000/yr

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← Meaningful heterogeneity
in some dimensions

Estimates of Risk Aversion

Risk Preference Analysis

	Absolute Risk Aversion	Interpretation
Normal Heterogeneity		
Mean / Median Individual	$4.22 * 10^{-4}$	94.6
25th percentile	$2.95 * 10^{-4}$	96.1
75th percentile	$5.49 * 10^{-4}$	93.8
95th percentile	$7.31 * 10^{-4}$	92.2
99th percentile	$8.59 * 10^{-4}$	91.8

Literal (hard to interpret)
parameter

Value X that would make this individual
Indifferent between:

- (1) 50-50 chance of either winning \$100 or losing \$X
OR
(2) inaction/doing nothing

So $X=100 \rightarrow$ risk neutral, $X < 100 \rightarrow$ risk averse

$\rightarrow$ Not a lot of heterogeneity

Counterfactuals: Reducing Switching Costs

- Motivation:
 - Better match of product to consumer increases consumer surplus
 - Makes market more competitive
 - But...may exacerbate adverse selection and reduce consumer surplus
- A “nudge”
 - Scale down switching cost to factor Z
 - So falls from $\$ \eta$ to $\$ Z \eta$

Consumer Surplus Changes

- Calculate as Certainty Equivalent
 - Money Q to make indifferent between Q and the risky good j

$$u(Q_{kjt}^{\kappa}) = -\frac{1}{\gamma_k(X_k^A)} e^{-\gamma_k(X_k^A)(W-Q_{kjt}^{\kappa})} = U_{kjt}(P_{kjt}, \kappa Z \eta_k, \mathbf{1}_{kj,t-1})$$

CTF: Partial Equilibrium (no supply-side adjustment)

t_1

t_2

Policy: reduce switching costs to 0.25η

Mean Welfare Change: % Total Emp. Spending

Mean Total Emp. Spending	\$4,373	\$4,486
Welfare Change Population	2.2%	2.5%
Welfare Change Switchers	4.0%	4.4%

Non-switchers have no welfare change

CTF: Full Equilibrium (supply-side/price adjusts)

Full Equilibrium Welfare Analysis Information Provision: η to $.25\eta$

	t_1	t_2	t_4	t_6	Avg. t_1-t_6
Welfare Change: % Total Spending					
Mean Total Emp. Spending	\$3,755	\$4,097	\$4,022	\$3,862	\$4,015
Welfare Change Population	-1.7%	-2.5%	-3.6%	-3.06%	-2.9%
Welfare Change Switchers	2.3%	4.3%	6.1%	6.3%	4.6%
Welfare Change Non-Switchers	-5.5%	-9.5%	-13.8%	-11.2%	-11.0%

Intuition: healthy enrollees in comprehensive plan will switch, which raises the costs/premiums of remaining enrollees

Surprising result! : helping consumers improve their choices reduces total consumers surplus (though it helps some)

Further Comments on Supply Side

- This assumes “competitive” pricing
- In oligopoly setting, how should a firm price given switching costs?
 - See Farrell & Klemperer (2007) for theory
 - “competitive” institutionally matches this firm’s health plan, but could imagine other settings

Potential “Enrichment” of Model

- Unobserved shocks for each individual are correlated over time
 - Otherwise, tend to overstate switching costs
If unobserved plan preferences, nudges won't work
 - See Dube, Hitsch, Rossi for possible model
- Dynamic model of consumer choice – consumers “rationally” chose knowing existence of switching costs
 - Realistic? Or are consumers myopic?
 - Shcherbakov (2007)

Question for Handel: Is switching cost heterogeneity correlated with health status?

- What if customers with more experience with their health care plan (i.e. “sicker” enrollees) have lower switching costs?
 - Then less potential to nudge “sick” patients
 - Then nudging may have little impact on enrollees on comprehensive plan
 - ➔ Handel may overstate how much nudging reduces consumer surplus?

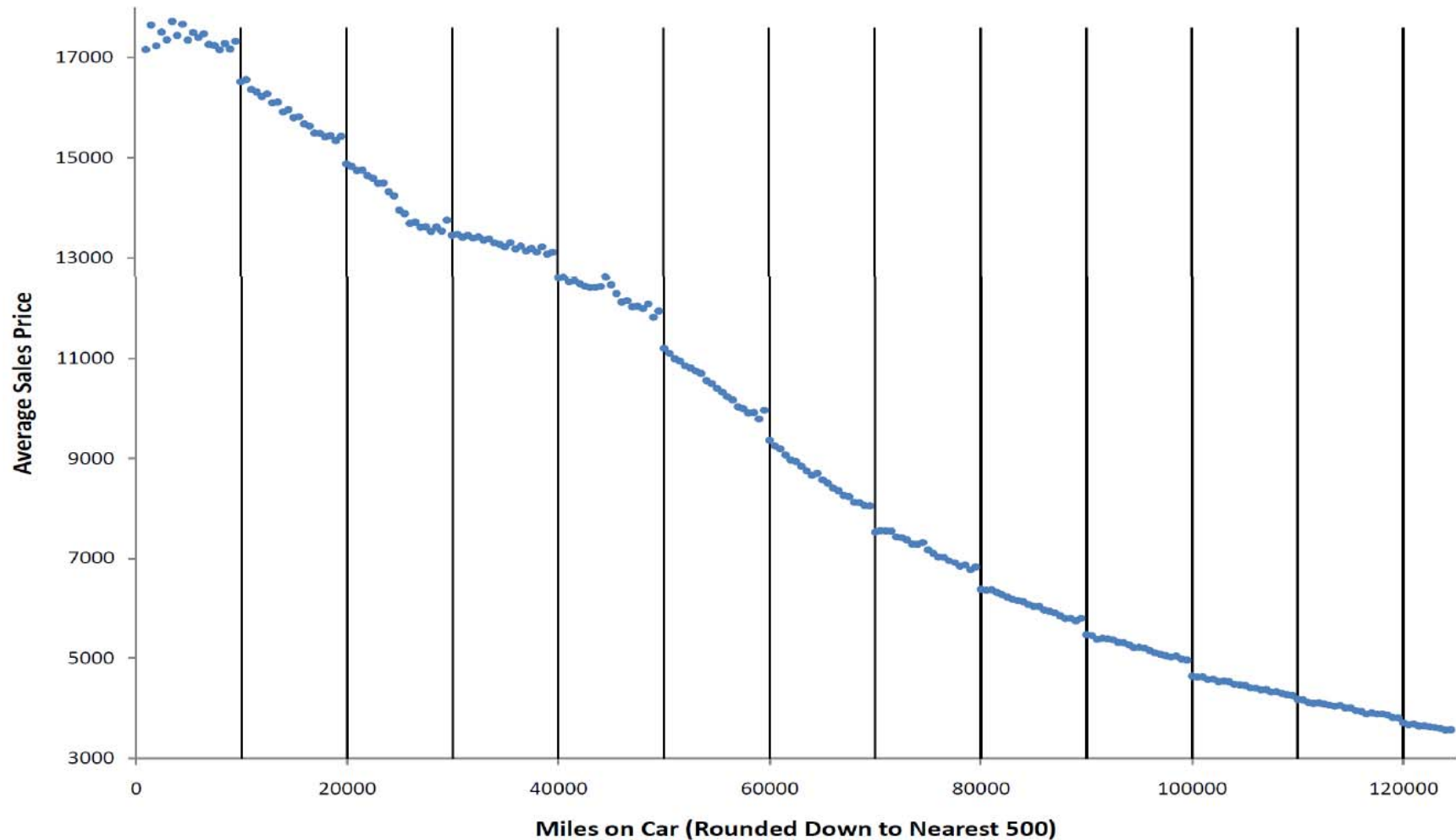
Other Work on “Inattentive Consumers”

- See various papers from 2010 FTC Microeconomics conference
 - <http://www.ftc.gov/be/workshops/microeconomics/agenda.shtm>
- Good overview David Laibson
 - http://www.ftc.gov/be/workshops/microeconomics/2010/docs/laibson_slide.pdf
- Examples below
 - Mileage on car – Lacetera, Pope and Sydnor (2010)
 - Bank overdraft fees – consumers pay attention after fees but then forget – Stango and Zinman (2010)
 - Financial decisionmaking as a function of age – Laibson & co-authors

The End

“Left Digit” Bias

Results – Average price of cars sold at auction by mileage

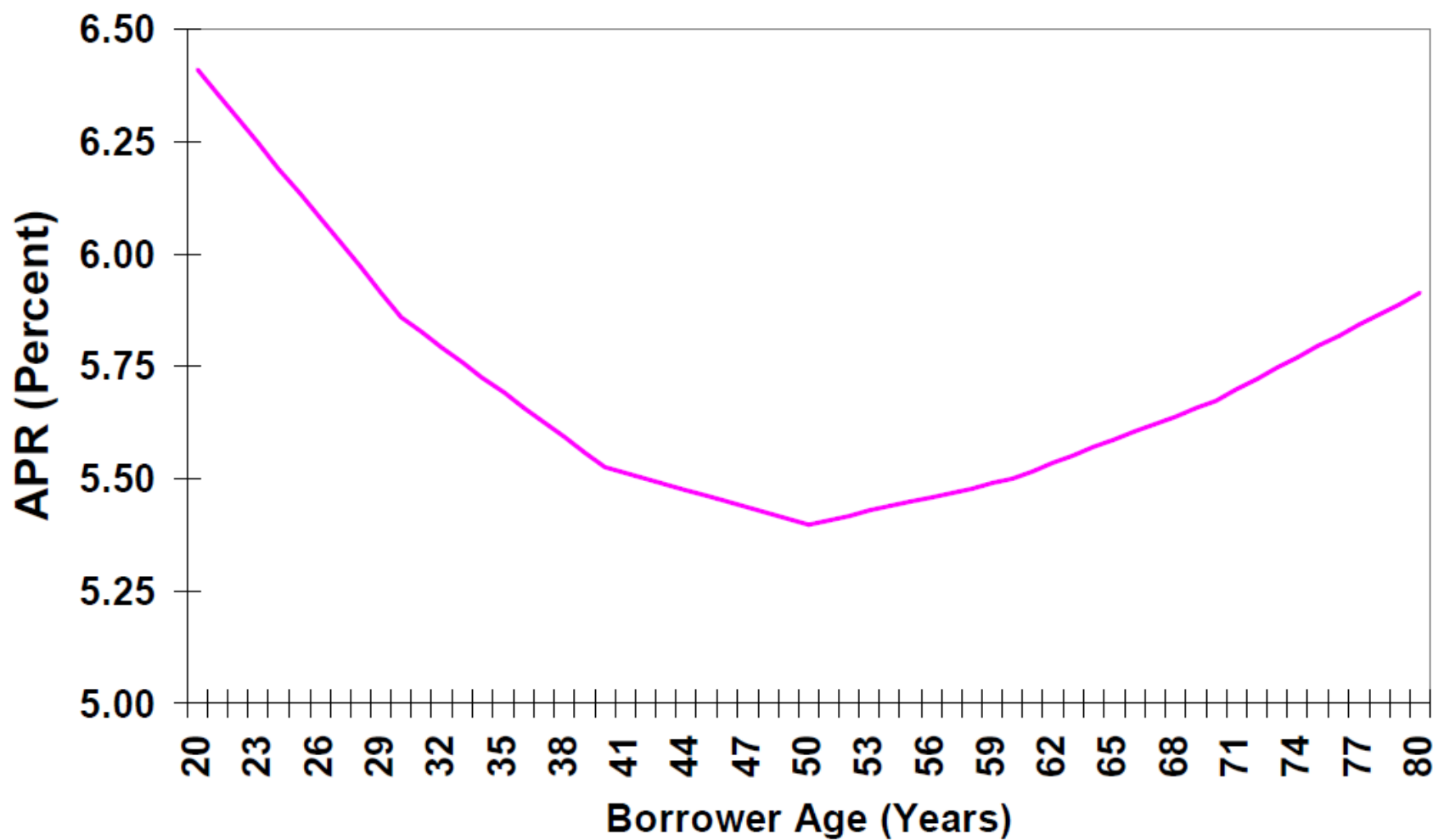


In separate Canadian data, discontinuities exist at 10,000 Kilometer marks

Consumer “Limited Attention” and Salience

- Overdraft fees on checking account
 - More spending than on large appliances
- Use surveys of checking customers about their accounts
 - Questions about OD: “Do you have overdraft protection?”
 - Non-OD questions → falsification tests
- Find both immediate and medium-run reductions in overdrafts following OD surveys (but not non-OD surveys)
 - Stronger effects for customers with low education; no difference by income

Home Equity Loan APR by Borrower Age



See Laibson & co-authors

Other Evidence of Consumer Mistakes

- The power of the default choice
 - Madrian & Shea (2001) 401(k) participation higher when enrollment is automatic
 - Duarte & Hastings
 - Hortacsu, Madanizadeh & Puller
- Analytic cognitive function declines with age